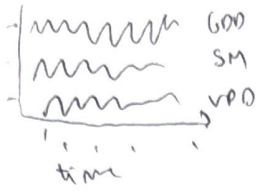


A. A NEW APPROACH

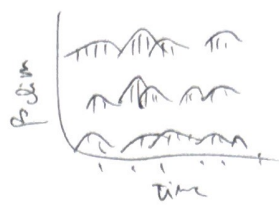
Individual trends



+ climate predictors with 7 species flat



B. WITH SPECIES STUFF (because I'm an ecologist, I mean)



SPERMATOPHYTES? ABOUT 3 SPECIES?



ONE TREE!



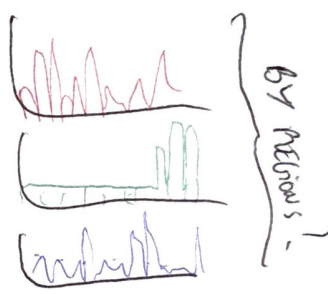
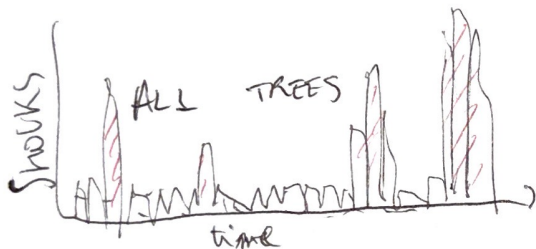
REGIONS

EXTREME YEARS!

Somewhat concordant



PARADOXICAL SHOCKS

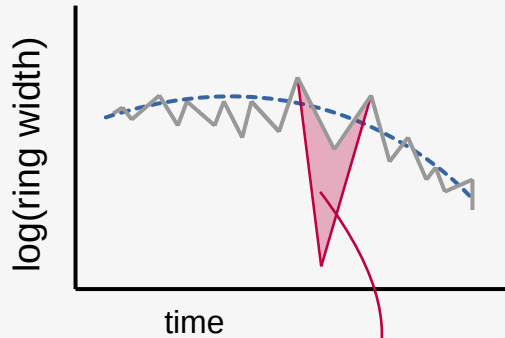
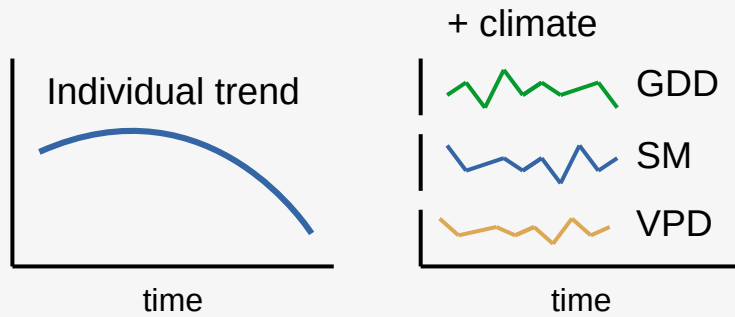


BY REGIONS?

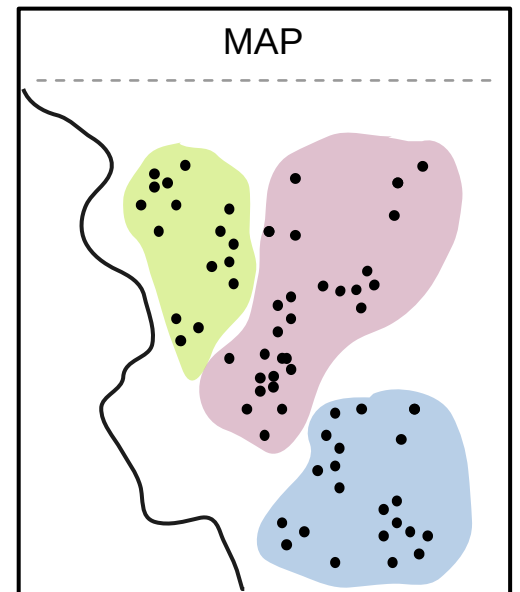
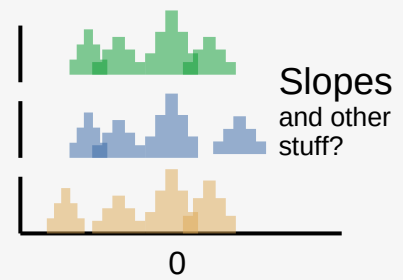
NOT PREDICTED BY CLIMATE! + TAKE-NOTE?

?

A. New approach...

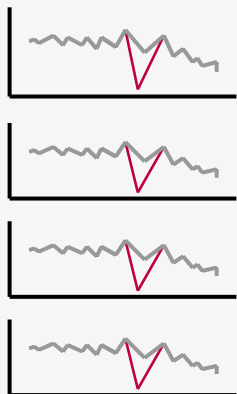


B. Species sensitivity



C. Extreme years!

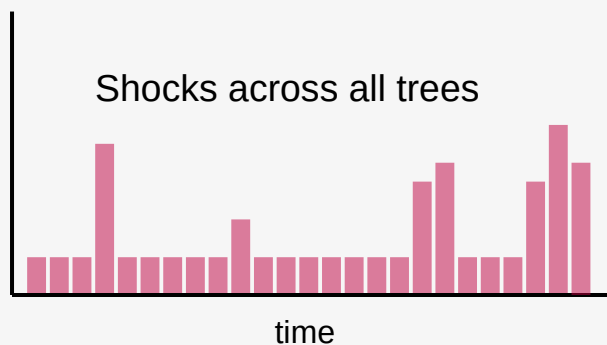
Concordant signal



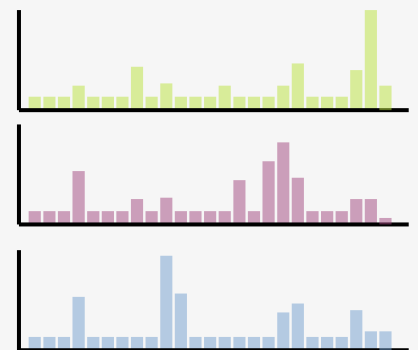
Shock parameter 1

Shock parameter 2

Shocks across all trees

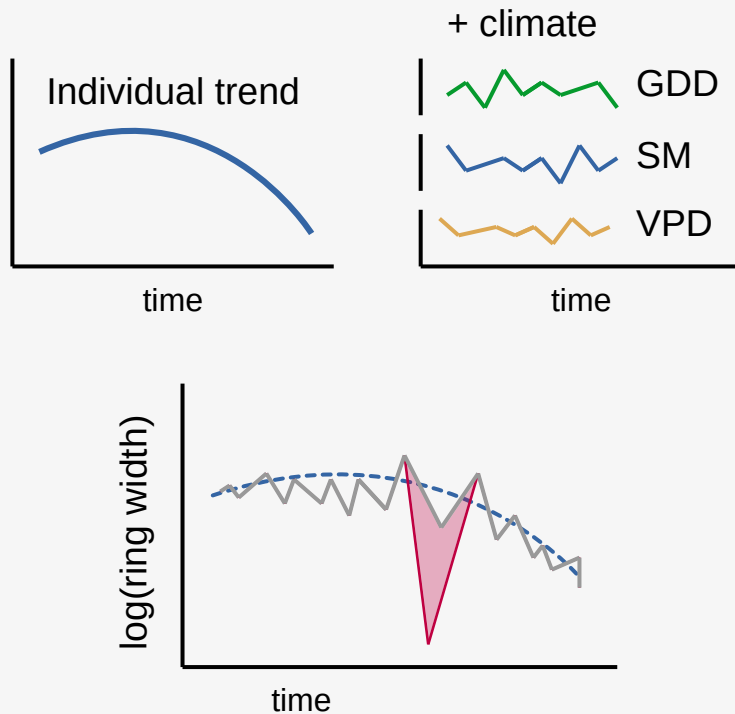


Regions



D. Not predicted by climate... +take-home message?

A. New approach...

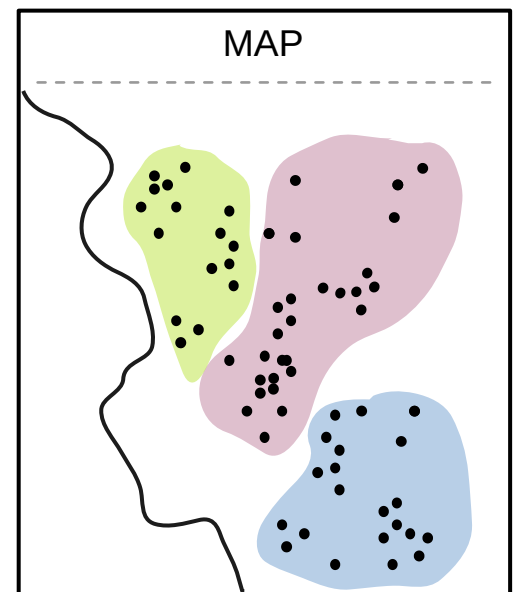
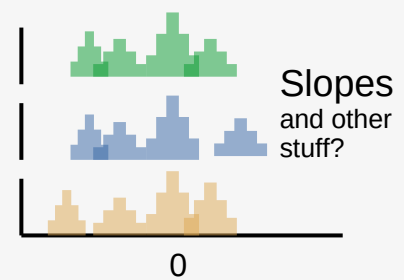


“We developed a new modeling framework that integrates individual-level growth trends with species-specific climate sensitivities”

“This hierarchical Bayesian model [...] is specifically designed to capture extreme growth responses across trees, species and ecosystems”

“we apply the model to 1.6 million observations of tree-ring width across Western North America”

B. Species sensitivity

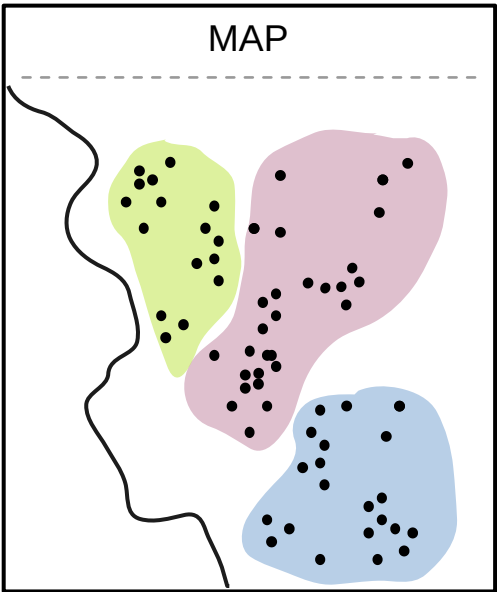


Regions

“We identify the significant drought period of the 2000-2007 as causing exceptional reductions in tree growth.”

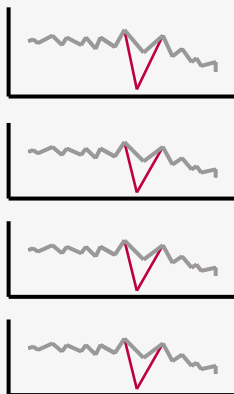
“These reductions are well beyond those predicted from direct responses to temperature, vapor pressure deficit and soil moisture or their interactions, suggesting gaps in our fundamental understanding of tree growth responses to climate”

“Ultimately, these results demonstrate that tree growth is a critical indicator of drought, and that many current models may underestimate growth declines associated with extreme drought events.”



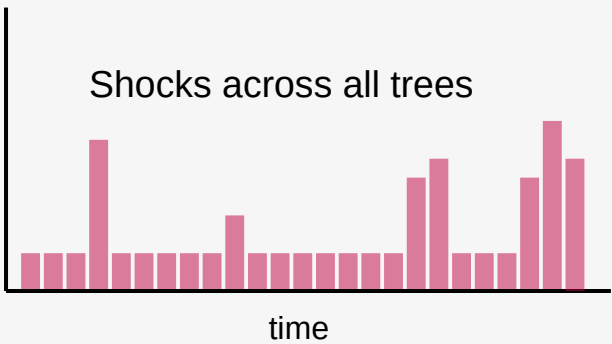
C. Extreme years!

Concordant signal

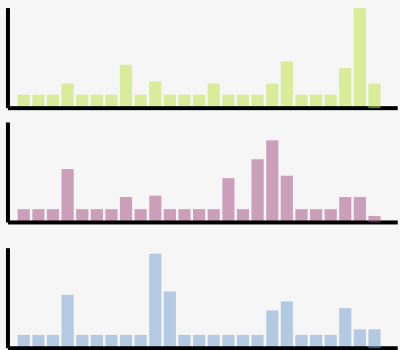


Shock
parameter 1

Shock
parameter 2



Regions



D. Not predicted by climate... +take-home message?